AP PRECALCULUS

Unit 1 • Worksheet 1.7-1.11A • Answer key

Review 1.7-1.11A • Review of Topics 1.7 to 1.11

Answer key with worked solutions

Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1 (1.7)

NO CALCULATOR

Which line is a horizontal asymptote of $r(x) = \dfrac{3x^2 - x}{6x^2 + 5}$?

(A) $y = 3$
(B) $y = 0$
(C) $y = -\frac{1}{5}$
(D) $y = \frac{1}{2}$

Solution

Answer: (D). $\frac{3}{6}$.

Question 2 (1.9)

NO CALCULATOR

What is $\lim\limits_{x \to 2^+} \dfrac{x+1}{2-x}$?

(A) 0
(B) 3
(C) ∞
(D) $-\infty$

Solution

Answer: (D). Denominator negative near 2^+.

Question 3 (1.10)

NO CALCULATOR

Let $r(x) = \dfrac{x^2 - 1}{x^2 + x - 2}$. What is true at $x = 1$?

(A) vertical asymptote
(B) hole at $\left(1, \frac{2}{3}\right)$
(C) zero
(D) hole at $(1, 0)$

Solution

Answer: (B). Cancel $(x-1)$; $\frac{2}{3}$.

Question 4 (1.8) NO CALCULATOR

Which value is a zero of $r(x) = \dfrac{(x+1)(x-5)}{(x-5)(x+3)}$?

(A) -1 only
(B) -1 and 5
(C) 5 only
(D) -3

Solution
Answer: (A). 5 is excluded.

Question 5 (Free response • AP style) NO CALCULATOR

Let $r(x) = \dfrac{2x^2+2x-4}{x^2-1}$.

(A) (i) Find the zeros of r and any hole. (ii) Find the vertical asymptote with limit statements.
(B) (i) Write the end behavior with a limit. (ii) Solve $r(x) \geq 0$.

Solution
(A)(i) Zero -2; hole $(1,3)$. (ii) $x=-1$: $\lim_{x\to -1^-} = -\infty$, $\lim_{x\to -1^+} = +\infty$.
(B)(i) $\lim_{x\to\pm\infty} r = 2$. (ii) $(-\infty,-2]\cup(-1,1)\cup(1,\infty)$.